

Question number	Answer	Additional guidance	Mark
5(a)(i)	A description that includes the following points: - as light intensity increases pupil diameter decreases (1) - LED lights cause a greater reduction in pupil diameter than incandescent light (1)	ALLOW negative correlation ALLOW explained using data from table	Graduate (2)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An answer showing the following steps: - find the area of the pupil in (mm²) (1) - convert to m² (1) - multiple the intensity by the area (1)	Example of calculation: $A = \pi r^2 = 3.14 \times 8^2 = 201 \text{ mm}^2$ $A = 0.000201 \text{ m}^2$ $\text{Answer} = 0.000201 \times 1400 = 0.2814/0.28/0.281$ ALLOW 28/281/2814 for 2mps Correct answer 3 marks	Expert (3)

Question number	Answer	Additional guidance	Mark
5(a)(iii)	A description that includes the following points: • light is detected by {rods / cones} in the retina (1) • reflex arc formed from {sensory neurones, relay neurones and motor neurones} (1) • (transmit) impulses to muscles in the iris (1) • radial muscles relax and circular muscles contract constricting the pupil (1)	ALLOW effect on rhodopsin described. ALLOW reflex pathway correctly described	Expert (4)

Question number	Answer	Additional guidance	Mark
5(b)(i)	An answer that includes the following points: • {self-renewing/ continuously dividing} cell (1) • that can give rise to {many / most} of the different cell types (1)	IGNORE Hayflick limit ALLOW cell that can undergo mitosis IGNORE can give rise to all ALLOW all cell types apart from extra embryonic tissue	Expert (2)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An answer that includes the following points: • stem cells could continue to divide (1) • stem cells could differentiate into other cell types (1)	ALLOW could form a tumor / a cancer	Expert (2)

